Muhammad Hasnain

GeoAI Developer & Meteorologist

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GIS Analyst, Meteorologist and GIS Developer with five self-conducted research projects at national level. Skilled in cartography, climate modeling and geospatial analysis. Experienced in executing projects with organizations such as Geo Mapping Solutions, the Pakistan Meteorological Department and the Senate of Pakistan (Parliament House). Proficient in ArcGIS Pro, QGIS, DSSAT, Python and Google Earth Engine, with a strong focus on solving environmental challenges and real-world problems through data-driven insights. Expanding into Machine Learning/AI, LLM-based tools and Agentic AI engineering as a Climate Modeler, and skilled in frontend and backend development, including building a web portal for NDMA.

Experience

Jul 2026 – present

Employee, full-time
National Disaster Management Authority (NDMA) Pakistan, Islamabad

- Building the NDMA Cloud-based Early Warning System: forecasts from seven weather models, flood, drought, crop water and heatwave modules on one map, with a FastAPI backend and an AI briefing assistant.
- Built Heatwave Intelligence, a standalone portal with 45 years of PMD station data, trend analysis, a 7-day outlook and a chat agent in English and Roman Urdu.
- Bias-corrected the 7-day heatwave outlook on 111 PMD stations; tested on July–August 2026, it caught 80% of observed heatwave days.

Jan – Jun 2026

Intern
Senate of Pakistan, Parliament House, Islamabad

- Developed a Python model that automates payment processing in place of manual Excel work.
- Reduced calculation time from hours to seconds and improved accuracy in financial operations.

Jul 2025 – present

Image Annotator, part-time
TerraSkyAI Inc., remote

- Labelling and annotating image data used to train and evaluate AI/ML computer vision models.

2025, 3 months

Meteorology Intern
Pakistan Meteorological Department, Islamabad

- Developed an integrated Python model for drought mapping across Pakistan from several drought indices.
- Automated drought analysis that was previously calculated by hand.

2024, 3 months

Meteorology Intern
Pakistan Meteorological Department, Islamabad

- Analyzed satellite data for climate forecasting and extreme weather reports.
- Developed an Android application that helps farmers analyze crop patterns.
- Developed an integrated model that predicts rainfall patterns using Python and forecasting models.

2023, 1 year

GIS Analyst
Geo Mapping Solutions Pvt Ltd, Islamabad

- Multispectral and hyperspectral image processing; remote sensing data interpretation.
- Digitization of parcel-level vector layers; georeferencing of Mussavi sheets on satellite imagery.
- Geodatabase development, topological corrections and data validation, query-based spatial analysis.
- Cartographic and thematic map production; work plan implementation for projects.

Featured project

Heatwave Intelligence, a portal for NDMA

Shows how heat in Pakistan has changed since 1980 and which districts face a heatwave in the next seven days: PMD records 1980–2024 on a 0.45° grid, period and year comparison, Mann-Kendall trends, rankings and records, an ECMWF 7-day outlook bias-corrected to 111 stations, and a chat agent whose every number comes from a data tool. Screens at hasnain061-ai.github.io.

Geo-AI projects

- Heatwave Intelligence portal (agentic AI) for NDMA
- Flood Inundation Monitoring portal (agentic AI) for NDMA
- Drought Monitoring portal (agentic AI) for PMD
- Crop Water Requirement portal for NDMA
- Coastal Hazard portal for NDMA
- Pakistan Ocean Climate Portal (in progress)
- Real-estate site selection using machine learning
- Network analysis for an Ohio county and for Germany in Python

Research

- Localized changes in heatwave dynamics across Pakistan and large-scale climate patterns
- Assessing the role of heatwaves in triggering flash droughts across Pakistan using remote-sensing indicators
- Advanced integration of drought mapping using several drought indices across Pakistan
- Mapping and monitoring of LULC change and potential causal factors in the Potohar region, using multi-temporal satellite data and GIS
- Decision support system for farmers based on the agro-climatic regions of Pakistan (Android application)

Skills

Remote sensing and GIS
Google Earth Engine, ArcGIS Pro, QGIS, ENVI, Pix4Dmapper, cartography, GeoAI

Climate and weather
GrADS, CDO, DSSAT, climate modeling, climate data analytics, heatwave and drought indices

AI and machine learning
Agentic AI and agentic engineering, LLMs, machine learning, Roboflow, data annotation

Software
Python, R, FastAPI, Leaflet, Mapbox GL JS, frontend and backend development, Microsoft Office, technical writing

Education

COMSATS University Islamabad
Bachelor of GIS and Remote Sensing, Atmospheric Sciences and Meteorology, Oct 2022 – Jun 2026, CGPA 3.73

Certifications

Flood Hazard and Exposure Mapping UNESCO

Introduction to Machine Learning for Earth Observation EO College

Cartography Esri

Going Places with Spatial Analysis Esri

Strengthening Climate Adaptation and Resilience UNITAR, UN CC

Becoming a Climate Champion UNITAR

Career Essentials in Generative AI Microsoft and LinkedIn

Career Essentials in Data Analysis Microsoft and LinkedIn

Introduction to Artificial Intelligence LinkedIn Learning

Ortho-rectification of Aerial Imagery through Orthophoto.io Geo University

Workshops, volunteering and languages

- International DSSAT Workshop for Crop Modeling
- Disaster Management Training (early warning alerts)
- AGRISAT Satellite Data Training Workshop
- Volunteer: Pakistan Minerals Investment Forum 2025; Pakistan Learning Festival; HRCP; Federal Ministry of Education (DWE)
- English C1 (spoken B2); German A1

References available on request.